Performance Optimization of FAS-Assisted U2G Communications Using Reinforcement Learning

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***Abstract*—Fluid antenna systems (FAS) are emerging as a promising technology for sixth-generation (6G) networks due to their flexible and reconfigurable radiation characteristics. In this paper, we investigate the performance optimization of FAS-assisted unmanned aerial vehicle (UAV)-to-ground (U2G) communications. Unlike conventional static antenna configurations, the proposed framework leverages the unique ability of FAS to dynamically activate and adjust antenna ports, enabling optimization in U2G channels. A reinforcement learning approach based on an enhanced proximal policy optimization (PPO) algorithm, termed Memetic PPO, is introduced to efficiently explore the discrete FAS configuration space by integrating a local search mechanism. Key performance metrics, including channel capacity and modeling accuracy, are derived and analyzed. Simulation results demonstrate that the number of active ports and port spacing significantly affect both metrics, and the proposed Memetic PPO algorithm achieves superior configurations by effectively balancing these factors. These findings provide valuable insights for the design and optimization of FAS in practical UAV communication systems.**

***Index Terms*—Fluid antenna system, unmanned aerial vehicle, channel capacity, reinforcement learning, proximal policy optimization.**

I. Introduction

Despite the remarkable achievements of fifth-generation (5G), it still faces some shortcomings, such as an over-reliance on base stations and high costs. In contrast, sixth-generation (6G), as the next-generation communication technology, aims to integrate satellites to build a globally covered network and features low costs, holding promising development prospects. The fluid antenna system (FAS) is regarded as one of the key technologies for 6G and has demonstrated vigorous development momentum. The core concept of FAS lies in the reconfigurability of antenna shape and position, with similar ideas traceable back to the directionally reconfigurable waveguide antenna proposed in 1960. It was not until recent years that the concept of FAS was proposed in [1] and [2]. Essentially, FAS refers to any controllable fluid, conductive, or dielectric structure that can dynamically alter its shape and position to reconfigure key parameters such as gain, polarization, radiation pattern, and operating frequency, thereby enhancing signal transmission efficiency, spectrum utilization, and coverage. Currently, there exists substantial academic research on FAS. [3] indicated that under identical conditions, FAS exhibits higher secrecy capacity, lower outage probability, and superior energy efficiency compared to conventional antennas. Furthermore, several effective hardware implementations have emerged. For instance, the PB-FAS proposed in [4] offers low power consumption, low cost, and easy integration, making it suitable for large-scale smart city deployment. [5] found that in communication scenarios where base stations are equipped with fluid antennas, channel capacity can be enhanced by optimizing the antenna positions at both the base station and the user ends. [6] pointed out that when users are equipped with FAS, the scalability of the network can be improved by alleviating the base station's channel state information burden. [7] comprehensively presented the design, channel modeling, and experiments of an ER-FAS, the results demonstrated that their proposed ER-FAS performs effectively in both the near-field and far-field, thereby enhancing the overall spectral efficiency of the wireless communication system. Recent years have witnessed the rapid development of artificial intelligence (AI), and the deep integration of communication and AI aligns with the vision of 6G. [8] first introduced Bayesian regression into FAS and proposed S-BAR, a general, high-precision, low-pilot-overhead channel estimation method for FAS with strong robustness and adaptability.

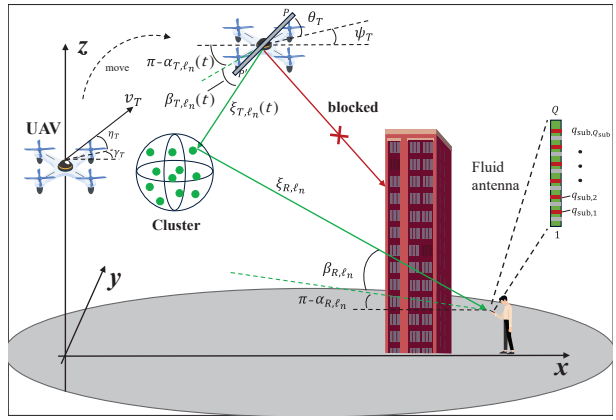

Fig. 1: Channel model of UAV-to-ground communication scenario assisted by FAS.

Although research on FAS is extensive, applying reinforcement learning (RL) to dynamic unmanned aerial vehicle (UAV)-to-ground (U2G) scenarios remains limited [8], [9]. To address the challenges of time-varying channels and the low exploration efficiency of baseline proximal policy optimization (PPO) in discrete action spaces, we propose an enhanced variant termed *Memetic PPO*. This algorithm

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integrates a local search mechanism to refine action selection before execution, significantly accelerating convergence and uncovering superior configurations. Furthermore, considering the model error discussed in [10], our framework optimizes FAS configurations by balancing the trade-off between channel capacity and modeling accuracy, providing better solutions for practical UAV communication systems.

Notation: The lowercase, boldface lowercase, and boldface uppercase letters such as x, $\mathbf{x}$, and $\mathbf{X}$ stand for the scalars, vectors, and matries, respectively. $\|\cdot\|$ represents the Frobenius norm, $\|\cdot\|_\infty$ represents the ℓ_∞ norm and $(\cdot)^{\mathrm{H}}$ denotes the conjugate transpose operation.

II. System Model

Consider a three-dimensional (3D) channel model for A2G communications as shown in Fig. 1, the UAV moves at a speed of v_T, and the azimuth and elevation angles of the direction of the motion are γ_T and η_T, respectively. In our model, the projection of midpoint of the UAV on the ground is defined as the coordinate origin, and the direction pointing to the receiver on the ground is the positive direction of the x-axis; the z-axis goes straight up and y-axis follows the right-hand rule. It is worth mentioning that the coordinate system does not move along with the UAV. Define H_0 and D_0 as the initial altitude of the UAV and the distance from the origin to the receiver, respectively.

On the UAV end, the azimuth and elevation orientation angles of the P omni-directional uniform linear array (ULA) antennas equipped on the UAV are respectively described as ψ_T and θ_T. The position of the p-th element in ULA can be expressed as

$$\mathbf{d}_p(t) = \mathbf{d}_T(t) + \mathbf{d}_{T,p}, \tag{1}$$

where $\mathbf{d}_T(t)$ denotes the time-varying position of the UAV, and $\mathbf{d}_{T,p}$ denotes the distance vector from the ULA midpoint to the p-th antenna. To be specific, they can respectively be expressed as

$$\mathbf{d}_T(t) = v_T t \begin{bmatrix} \cos\eta_T \cos\gamma_T \\ \cos\eta_T \sin\gamma_T \\ \sin\eta_T \end{bmatrix} + \mathbf{d}_{t_0}, \tag{2}$$

$$\mathbf{d}_{T,p} = k_p \delta_T \begin{bmatrix} \cos\theta_T \cos\psi_T \\ \cos\theta_T \sin\psi_T \\ \sin\theta_T \end{bmatrix}, \tag{3}$$

where $k_p = (P - 2p + 1)/2$, $\mathbf{d}_{t_0} = [0, 0, H_0]^{\mathrm{T}}$ and δ_T is the distance between adjacent antennas.

The receiver on the ground is equipped with a fluid antenna with a length of $W\lambda$ and Q ports, where λ is the carrier wavelength and $Q \gg Q_{\text{sub}}$, with Q_{sub} being the number of activated ports. The adjacent ports spacing of the fluid antenna Δ can be expressed as

$$\Delta = \frac{W\lambda}{Q-1}. \tag{4}$$

To characterize the dynamic port activation, we define a mapping relationship between the activated port index and the physical port index. Specifically, the q_{sub}-th activated port ($q_{\text{sub}} = 1, 2, \ldots, Q_{\text{sub}}$) corresponds to the q-th physical port ($q = 1, 2, \ldots, Q$) in the FAS through a mapping function $q = \mathcal{M}(q_{\text{sub}})$. Therefore, the position of the q_{sub}-th activated port relative to the global coordinate origin can be expressed as

$$\mathbf{d}_q = \mathbf{d}_R + k_q \Delta \begin{bmatrix} \cos\theta_R \cos\psi_R \\ \cos\theta_R \sin\psi_R \\ \sin\theta_R \end{bmatrix}, \tag{5}$$

where $k_q = (Q - 2\mathcal{M}(q_{\text{sub}}) + 1)/2$, ψ_R, θ_R are respectively the azimuth and elevation orientation angles of the FAS, and $\mathbf{d}_R = [D_0, 0, 0]^{\mathrm{T}}$ accounts for the position of receiver relative to the global coordinate origin.

The signal transmitted from the UAV will go through non-line of sight (NLoS) components, where the signal undergoes scattering by the cluster before arriving at the receiver. Due to the mobility of the UAV, the departure angles from the scatterers become time-varying. For a fixed scatterer position $\mathbf{d}_{\ell n} = [x_{\ell n}, y_{\ell n}, z_{\ell n}]^{\mathrm{T}}$ in the global coordinate system, the relative position vector from the UAV to the scatterer is given by

$$\mathbf{d}_{T,\ell n}(t) = \mathbf{d}_{\ell n} - \mathbf{d}_T(t), \tag{6}$$

the time-varying azimuth and elevation angles of departure (AoD) are then computed as

$$\alpha_{T,\ell n}(t) = \arctan2\left(y_{\ell n} - [\mathbf{d}_T(t)]_y,\ x_{\ell n} - [\mathbf{d}_T(t)]_x\right), \tag{7}$$

$$\beta_{T,\ell n}(t) = \arcsin\left(\frac{z_{\ell n} - [\mathbf{d}_T(t)]_z}{\xi_{T,\ell n}(t)}\right), \tag{8}$$

where $\arctan2(\cdot,\cdot)$ denotes the four-quadrant inverse tangent function, and $\xi_{T,\ell n}(t) = \|\mathbf{d}_{T,\ell n}(t)\|$ is the Euclidean distance between the UAV and the scatterer, given by

$$\xi_{T,\ell n}(t) = \sqrt{\sum_{i\in\{x,y,z\}} \left([\mathbf{d}_{\ell n}]_i - [\mathbf{d}_T(t)]_i\right)^2}, \tag{9}$$

here, $[\cdot]_i$ denotes the i-th component of the vector in the global coordinate system.

We can represent the channel matrix for the proposed channel model as $\mathbf{H}^{\text{cluster}}(t,\tau)$, where τ denotes the path delay. Each element $h_{p,q_{\text{sub}}}(t,\tau)$ in $\mathbf{H}^{\text{cluster}}(t,\tau)$ stands for the complex channel impulse response (CIR) of the propagation link from the p-th element in ULA to the q_{sub}-th element in FAS, which can be given as

$$h_{p,q_{\text{sub}}}(t,\tau) = h^{\text{cluster}}_{p,q_{\text{sub}}}(t)\delta\left(\tau - \tau^{\text{cluster}}(t)\right), \tag{10}$$

where $\tau^{\text{cluster}}(t)$ is the path delay for the NLoS propagation link, and $h^{\text{cluster}}_{p,q_{\text{sub}}}(t)$ can be expressed as

$$\begin{aligned} h^{\text{cluster}}_{p,q_{\text{sub}}}(t) = \sum_{\ell\in L}\sum_{n=1}^{\ell_N} & e^{j\varphi_{\ell_n} - j\frac{2\pi}{\lambda}(\xi_{T,\ell_n}(t)+\xi_{R,\ell_n})} \\ & \times e^{j\frac{2\pi}{\lambda}k_p\delta_T\cos(\alpha_{T,\ell_n}(t)-\psi_T)\cos\beta_{T,\ell_n}(t)\cos\theta_T} \\ & \times e^{j\frac{2\pi}{\lambda}k_p\delta_T\sin\beta_{T,\ell_n}(t)\sin\theta_T} \\ & \times e^{j\frac{2\pi}{\lambda}k_q\Delta\cos(\alpha_{R,\ell_n}-\psi_R)\cos\beta_{R,\ell_n}\cos\theta_R} \\ & \times e^{j\frac{2\pi}{\lambda}k_q\Delta\sin\beta_{R,\ell_n}\sin\theta_R} \\ & \times e^{j\frac{2\pi}{\lambda}v_T t\cos(\alpha_{T,\ell_n}(t)-\gamma_T)\cos\beta_{T,\ell_n}(t)\cos\eta_T} \\ & \times e^{j\frac{2\pi}{\lambda}v_T t\sin\beta_{T,\ell_n}(t)\sin\eta_T}, \end{aligned} \tag{11}$$

where $\{\varphi_{\ell n}\}_{n=1,2,\ldots,N}$ represents the random phase, assumed to be independent and uniformly distributed as $\varphi_{\ell n} \sim \mathrm{U}[-\pi,\pi)$. The terms $\xi_{R,\ell n} = \|\mathbf{d}_{\ell_n} - \mathbf{d}_R\|$ denote distances between the receiver and the n-th path in the ℓ-th cluster. Furthermore, α_{R,ℓ_n} and β_{R,ℓ_n} are angles of arrival. The scatterer positions follow the von Mises distribution

$$f(\alpha) = \frac{e^{\kappa\cos(\alpha-\mu_\alpha)}}{2\pi I_0(\kappa)}, \tag{12}$$

where κ is the concentration parameter, μ_α is the mean value of α, and $I_0(\cdot)$ is the modified Bessel function of order 0.

For U2G wireless communication systems, channel capacity is a crucial indicator of the achievable spectral efficiency, which is given by

$$C = \log_2\left(\det\left(\mathbf{I}_{Q_{\text{sub}}} + \frac{\rho_{\text{SNR}}}{P}\overline{\mathbf{H}^{\text{cluster}}}(t,\tau)\overline{\mathbf{H}^{\text{cluster}}}^{\text{H}}(t,\tau)\right)\right), \tag{13}$$

where $\mathbf{I}_{Q_{\text{sub}}}$ denotes the identity matrix of size $Q_{\text{sub}} \times Q_{\text{sub}}$, ρ_{SNR} is the signal-to-noise ratio (SNR), and $\overline{\mathbf{H}^{\text{cluster}}}(t,\tau)$ represents the normalized channel matrix. It is worth mentioning that a fundamental characteristic of fluid antenna systems is the strong spatial correlation among densely distributed ports. Due to the compact and quasi-continuous structure of the fluid antenna, adjacent ports experience highly similar propagation paths, resulting in correlated channel responses.

III. Memetic PPO for Dynamic FAS Configuration

To address the dynamic FAS configuration problem in the time-varying U2G communication channel described in Section II, we model the task as a Markov Decision Process (MDP) and employ an enhanced PPO algorithm named memetic PPO.

A fundamental characteristic of fluid antenna systems is the strong spatial correlation among densely distributed ports. Due to the compact and quasi-continuous structure of the fluid antenna, adjacent ports experience highly similar propagation paths, resulting in correlated channel responses.

Mathematically, the spatial correlation between the q_i-th and q_j-th ports can be characterized as

$$R_{q_i,q_j} = \mathbb{E}[h_{q_i} h_{q_j}^*], \tag{14}$$

which primarily depends on the spatial separation between ports. Owing to the continuous geometry of the fluid antenna, this correlation can be approximated as a function of the inter-port distance, i.e.,

$$R_{i,j} \approx f(|q_i - q_j|), \tag{15}$$

where q_i and q_j denote the spatial positions of the corresponding ports.

In the proposed framework, such spatial correlation is inherently embedded in the channel model through shared scattering paths. Since all ports are influenced by the same set of propagation clusters while differing only in their spatial coordinates, the resulting channel matrix naturally exhibits structured correlation across adjacent ports. This implies that neighboring port configurations tend to yield similar performance, thereby forming a smooth optimization landscape.

Motivated by this observation, we incorporate a memetic local search mechanism into the PPO framework, enabling the agent to exploit local spatial continuity while maintaining global exploration capability.

The MDP is defined as $(\mathcal{S}, \mathcal{A}, R, \gamma)$. The state at time step t is given by

$$\mathbf{s}_t = [\mathbf{p}_t, \mathbf{v}_t, \tau_t, \mathbf{m}_t, \Delta H_t, C_{t-1}], \tag{16}$$

where $\mathbf{p}_t \in \mathbb{R}^3$ and $\mathbf{v}_t \in \mathbb{R}^3$ denote the UAV position and velocity, $\tau_t \in [0,1]$ represents the normalized time index, and $\mathbf{m}_t \in \{0,1\}^N$ is the binary mask of active ports. The term $\Delta H_t = \|\mathbf{H}_t - \mathbf{H}_{t-1}\|_F$ captures the temporal variation of the channel, while C_{t-1} provides historical performance information.

The action space consists of discrete operations that modify the port configuration, including activating an inactive port, deactivating an active port, shifting an active port to a nearby index, or maintaining the current configuration. These operations are constrained such that the number of active ports satisfies $K_{\min} \le K_t \le K_{\max}$. The reward function is designed to balance spectral efficiency and hardware cost, given by

$$r_t = r_{\text{base}} + \beta \cdot \max(0, C_t - C_{t-1}), \tag{17}$$

where $r_{\text{base}} = C_t/K_t$ denotes the port efficiency, and C_t and K_t represent the instantaneous capacity and the number of active ports, respectively.

The policy $\pi_\theta(a_t|s_t)$ is optimized using the PPO clipped surrogate objective

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{18}$$

where $\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$. The advantage $\hat{A}_t$ is computed using generalized advantage estimation

$$\hat{A}_t = \sum_{l=0}^{\infty}(\gamma\lambda)^l \delta_{t+l}, \tag{19}$$

with $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$. The value network is trained by minimizing

$$\mathcal{L}^{\text{value}} = \mathbb{E}_t\left[(V_\phi(s_t) - R_t)^2\right]. \tag{20}$$

To improve exploration efficiency in the large discrete action space, the memetic local search mechanism refines the action selection process. Given a candidate action $a_{\text{candidate}}$ sampled from the policy, a neighborhood $\mathcal{N}(a_{\text{candidate}})$ is constructed based on both action index proximity and physical adjacency of FAS ports. The final action is selected as

$$a_t^* = \arg\max_{a' \in \mathcal{N}(a_{\text{candidate}})}\left\{\frac{C(s_t,a')}{K(s_t,a')}\right\}. \tag{21}$$

This refinement is performed prior to interaction with the environment, ensuring that the executed action is locally optimal under the current channel realization.

The overall Memetic PPO procedure alternates between trajectory collection and parameter updates. At each time step, an action is first sampled from the policy and then refined via local search before execution. The resulting transition tuples are stored to compute advantage estimates and returns.

Algorithm 1 Memetic PPO for Dynamic FAS Configuration

Initialize policy parameters θ and value parameters ϕ
for each iteration **do**
  Initialize trajectory buffer $\mathcal{D}$
  for $t = 1, \ldots, T$ **do**
    Observe state s_t
    Sample action $a_{\text{candidate}} \sim \pi_\theta(\cdot|s_t)$
    Refine action:

$$a_t^* = \arg \max_{a' \in \mathcal{N}(a_{\text{candidate}})} r(s_t, a')$$

    Execute a_t^* and observe (s_{t+1}, r_t)
    Store (s_t, a_t^*, r_t) in $\mathcal{D}$
  end for
  Compute advantages $\hat{A}_t$ using GAE
  Update policy by maximizing $\mathcal{L}^{\text{PPO}}(\theta)$
  Update value network by minimizing $\mathcal{L}^{\text{value}}$
end for

Subsequently, multiple epochs of stochastic gradient ascent are performed to update the policy using the clipped objective, while the value network is updated via regression. By combining global policy optimization with local exploitation guided by physical channel structure, the proposed method significantly improves convergence speed and solution quality compared to conventional PPO in high-dimensional discrete FAS configuration problems.

IV. Results and Analysis

The experiment employs realistic parameters for UAV-to-ground communications: carrier frequency $f_c = 5$ GHz, UAV initial altitude $H_0 = 20$ m, initial horizontal distance $D_0 = 50$ m, number of UAV antennas $P = 4$, total FAS ports $Q = 30$, UAV antenna spacing $\delta_T = \lambda/2$, UAV azimuth orientation angle $\psi_T = \pi/2$, UAV elevation orientation angle $\theta_T = \pi/3$, UAV velocity $v_T = 3$ m/s, UAV motion azimuth angle $\gamma_T = \pi/3$, UAV motion elevation angle $\eta_T = \pi/3$, and fixed time instant $t = 2$ s. The scattering environment consists of $L = 3$ clusters with 10 paths per cluster, following a von Mises distribution with an environment factor $\kappa = 2.0$. Training utilizes the PPO algorithm with the following hyperparameters: learning rate 1×10^{-4}, steps per update 1024, batch size 128, epochs per update 4, discount factor $\gamma = 0.99$, generalized advantage estimation parameter $\lambda = 0.95$, clipping parameter $\epsilon = 0.2$, entropy coefficient 0.03, value function coefficient 0.5, and maximum gradient norm 1.0. The policy network comprises two hidden layers with 128 neurons each, using ReLU activation functions. For the memetic PPO, a local search radius of 1 is employed to refine actions before execution. Training proceeds for 20,480 timesteps, with evaluation performed every 2,048 timesteps using 16 episodes. The environment's maximum episode length is set to 64 steps to ensure efficient exploration. Model checkpoints are saved based on evaluation performance, and training progress is monitored through reward progression and modeling error metrics.

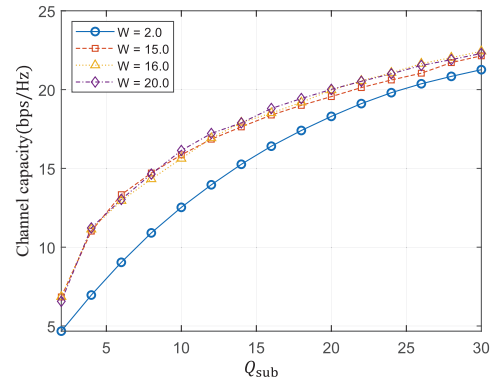

Fig. 2: Channel capacity of different FAS configurations

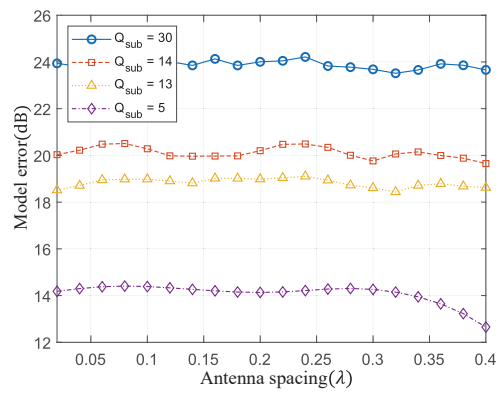

Fig. 3: Model errors with different FAS configurations

By setting $r_{\text{error}} = 0$ in (**??**), which corresponds to the scenario where the algorithm optimizes exclusively for channel capacity, the baseline PPO converges to an optimal FAS configuration of $W = 15.0$ and $Q_{\text{sub}} = 30$. In contrast, the memetic PPO yields a slightly different configuration of $W = 16.0$ and $Q_{\text{sub}} = 30$. The relationship between channel capacity and the parameters W and Q_{sub} is illustrated in Fig. 2. The simulation results reveal that for a fixed antenna length W, the channel capacity exhibits a significant upward trend as Q_{sub} increases. This phenomenon is primarily attributed to the fact that activating additional ports enhances the rank of the channel matrix $\mathbf{H}_{\text{FAS}}^{\text{cluster}}(t, \tau)$, which, according to (13), serves as a fundamental determinant of capacity. Specifically, the expanded spatial degrees of freedom facilitate richer multipath exploitation, thereby improving spectral efficiency. This observation is consistent with the findings in [10] and [11], corroborating the validity of our channel modeling and optimization results. Notably, while both algorithms identify the maximum number of ports ($Q_{\text{sub}} = 30$) as optimal, the memetic PPO attains a superior capacity by refining the antenna length to $W = 16.0$. This performance gain stems from the local search mechanism inherent in memetic PPO, which effectively explores the neighborhood of candidate actions to escape suboptimal solutions.

Correspondingly, by configuring $r_c = 0$ in (**??**), the optimization focuses exclusively on minimizing modeling error. Under this configuration, the baseline PPO identifies an optimal FAS setup of $W = 11.0$ and $Q_{\text{sub}} = 14$, while

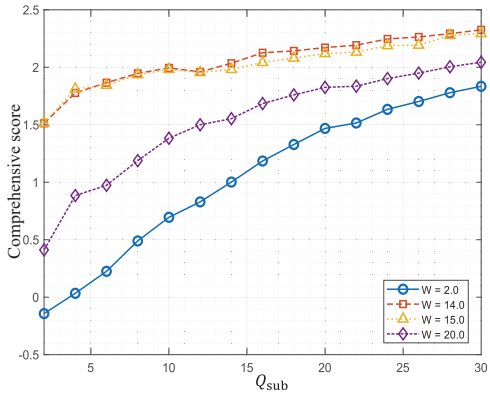

Fig. 4: Comprehensive score with different FAS configurations

memetic PPO refines this to $Q_{\text{sub}} = 13$. Fig. 3 illustrates the modeling error as a function of Q_{sub} and Δd for $W = 11.0$. Consistent with (**??**), the error increases monotonically with Q_{sub}. This occurs because denser port activation within a fixed spatial aperture exacerbates the deviation from the benchmark ULA channel characteristics. Notably, both variants converge to configurations with Q_{sub} values that are higher than the $Q_{\text{sub}} = 5$ configuration, which was observed in simulations to yield lower modeling error. This sub-optimality stems from two factors: (i) the non-stationary channel induced by UAV mobility creates a "narrow basin of attraction" for the rewards at extremely low Q_{sub}, making these optima difficult to sustain under stochastic exploration; and (ii) the PPO clipping mechanism, while ensuring stability, may bias the agent toward broader, more robust local optima that require less search pressure. Nevertheless, memetic PPO achieves a lower error than the baseline, demonstrating that its integrated local search effectively enhances exploration within the discrete action space to locate superior configurations.

To evaluate the system performance under a joint optimization framework, the weighting factors in (**??**) are empirically assigned as $r_c = 0.2$ and $r_{\text{error}} = -0.1$. These coefficients are normalized to enhance numerical stability during the training process, with the relative magnitudes reflecting a prioritized focus on maximizing channel capacity while simultaneously penalizing modeling inaccuracies. Under this multi-objective setting, the baseline PPO converges to a configuration of $W = 15.0$ and $Q_{\text{sub}} = 30$, whereas the memetic PPO identifies a more refined solution of $W = 14.0$ and $Q_{\text{sub}} = 30$. The functional relationship between the comprehensive score and the FAS parameters is illustrated in Fig. 4. The simulation results demonstrate that the composite utility increases monotonically with the number of activated ports. Although an increase in Q_{sub} inevitably exacerbates the modeling error [10], the chosen weighting strategy ensures that the gains in spectral efficiency dominate the overall performance metric. Specifically, the positive reward for capacity outweighs the penalty incurred by modeling deviations, thereby favoring configurations with high Q_{sub}. Furthermore, the configuration attained by the memetic PPO ($W = 14.0$) yields a superior comprehensive score compared to the baseline PPO's result ($W = 15.0$). This performance margin is attributable to the hybrid optimization architecture of memetic PPO, which synergistically combines stochastic gradient-based policy updates with deterministic local search to navigate the complex multi-objective landscape more effectively. The discrepancy in the identified optimal antenna length suggests that memetic PPO is capable of discovering a more balanced equilibriumłwhere a moderate antenna length maintains high capacity while effectively managing system complexity. This underscores the algorithm's robustness in capturing nuanced trade-offs that remain elusive to the standard PPO framework.

V. Conclusion

This paper has investigated the performance optimization problem for a FAS-assisted U2G communication system using reinforcement learning. The proposed framework can adaptively configure FAS parameters, such as antenna length and the number of active ports, in dynamic UAV scenarios, exploring the potential of data-driven methods for enhancing U2G communication performance. Through comprehensive simulation and comparative analysis, the effectiveness of the PPO algorithm and its memetic variant has been validated. It has been demonstrated that the number of active ports and the fluid antenna length significantly impact the trade-off between channel capacity and modeling error. Specifically, while increasing the number of active ports leads to higher channel capacity, it simultaneously elevates the modeling error, emphasizing the requirement for an intelligent configuration strategy. Moreover, the memetic PPO variant is shown to achieve superior comprehensive scores and better convergence stability compared to the standard PPO by integrating deterministic local search. Finally, it should be pointed out that applying this RL-based framework to multi-user interference scenarios and incorporating practical constraints like energy efficiency are key research directions in our future research.

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